

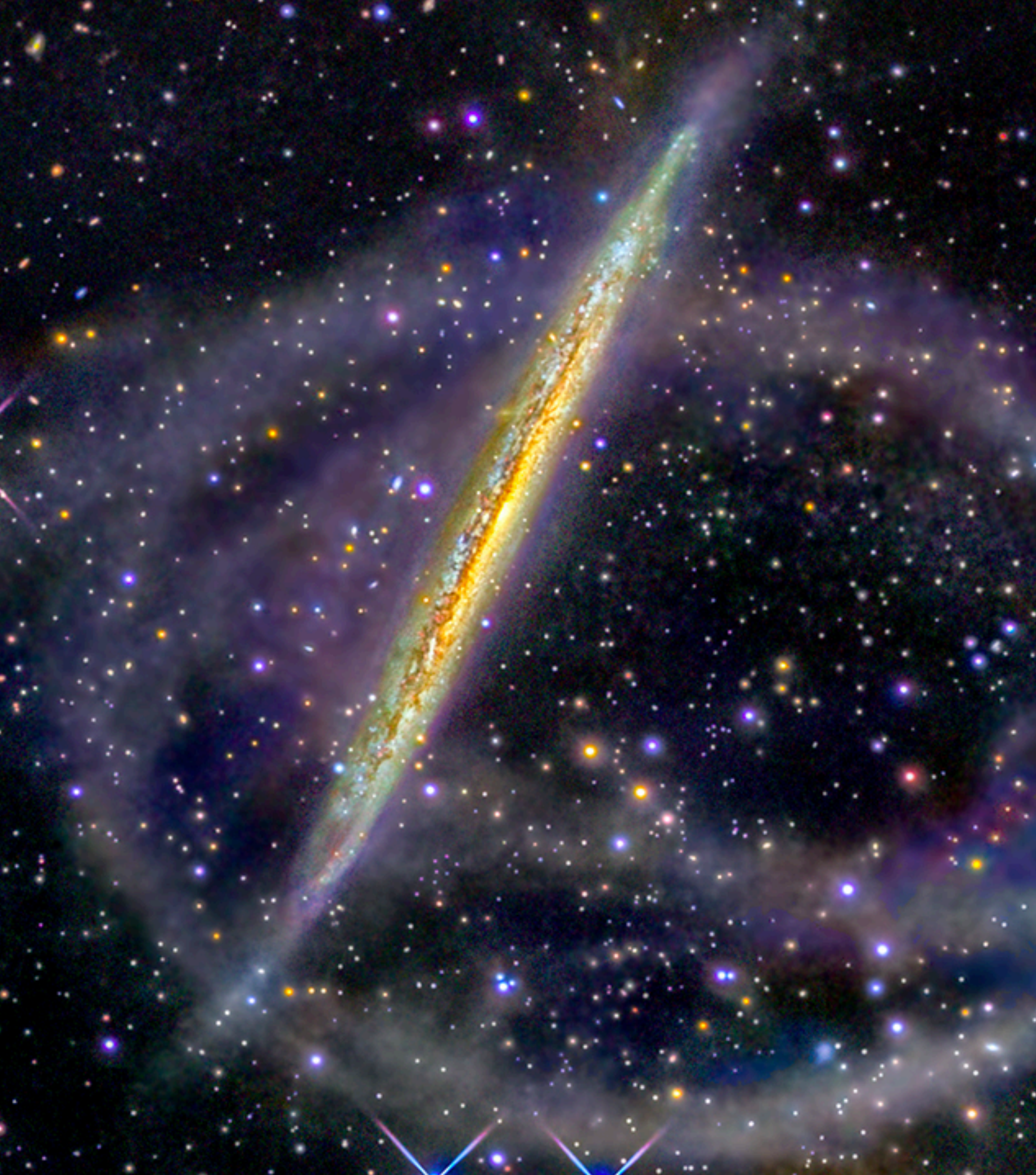
Galactic Dynamics

HW 7 is due next Thursday (11/30)

HW 8 is due 12/07 (last day of class)

Last lecture on Nov 28

100% project work time on Nov 30



Stars trace the density distribution of galaxies under two working assumptions:

- 1) the mass of the star(s) is much smaller than the mass of the galaxy
- 2) the “smoothing” length scale is much smaller than the size of the galaxy ($\ll 100$ kpc) but larger than the spacing between each star ($\gg 1$ pc)

The motion of the test mass particle is described by:

$$\boldsymbol{a}(\boldsymbol{x}) = -\nabla\Phi(\boldsymbol{x})$$

acceleration is the gradient of the gravitational potential

$$\nabla^2\Phi(\boldsymbol{x}) = 4\pi G \rho(\boldsymbol{x})$$

the potential is defined by the mass density of the galaxy

The simplest potential
is that of a point mass:

$$\Phi(r) = -\frac{G M}{r}$$

We can solve for the acceleration as:

$$\begin{aligned}\boldsymbol{a}(\boldsymbol{x}) &= -\nabla\Phi(\boldsymbol{x}) \\ &= -\frac{G M}{|\boldsymbol{x}|^3} \boldsymbol{x} \\ &= -\frac{G M}{r^2} \hat{\boldsymbol{x}}\end{aligned}$$

Once we have the acceleration, we can describe the motion of the test particle:

$$\begin{aligned}\ddot{\boldsymbol{x}} &= \boldsymbol{a}(\boldsymbol{x}) \\ \ddot{\boldsymbol{x}} &= -\nabla\Phi(\boldsymbol{x})\end{aligned}$$

“Kepler” equation:
solvable analytically for two body,
not analytically solvable generally

For the general case, it is useful to consider two coupled first-order differential equations that can be solved numerically

$$\begin{aligned}\dot{\boldsymbol{x}} &= \boldsymbol{v} \\ \dot{\boldsymbol{v}} &= -\nabla\Phi(\boldsymbol{x})\end{aligned}$$



$$\begin{aligned}\dot{\boldsymbol{w}} &= (\dot{\boldsymbol{x}}, \dot{\boldsymbol{v}}) \\ &= (\boldsymbol{v}, -\nabla\Phi(\boldsymbol{x}))\end{aligned}$$

where

$$\boldsymbol{w} = (\boldsymbol{x}, \boldsymbol{v})$$

We can write down how the test mass moves if we select initial conditions for the position and velocity. A commonly-used choice is the circular velocity:

Centripetal acceleration = gravitational acceleration

$$\frac{v^2}{r} = \left| \frac{d\Phi}{dr} \right|$$
$$v_{\text{circ}}(r) = \sqrt{r \left| \frac{d\Phi}{dr} \right|}$$

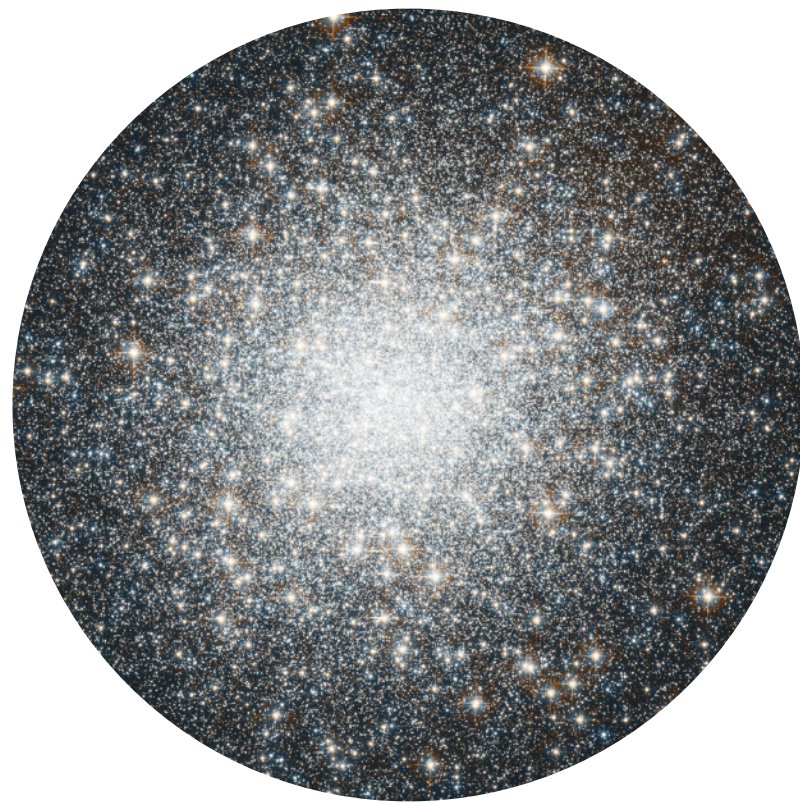
For the case of a point mass:

$$v_{\text{circ}}(r) = \sqrt{\frac{GM}{r}}$$

Extensions beyond a point mass:

Plummer sphere

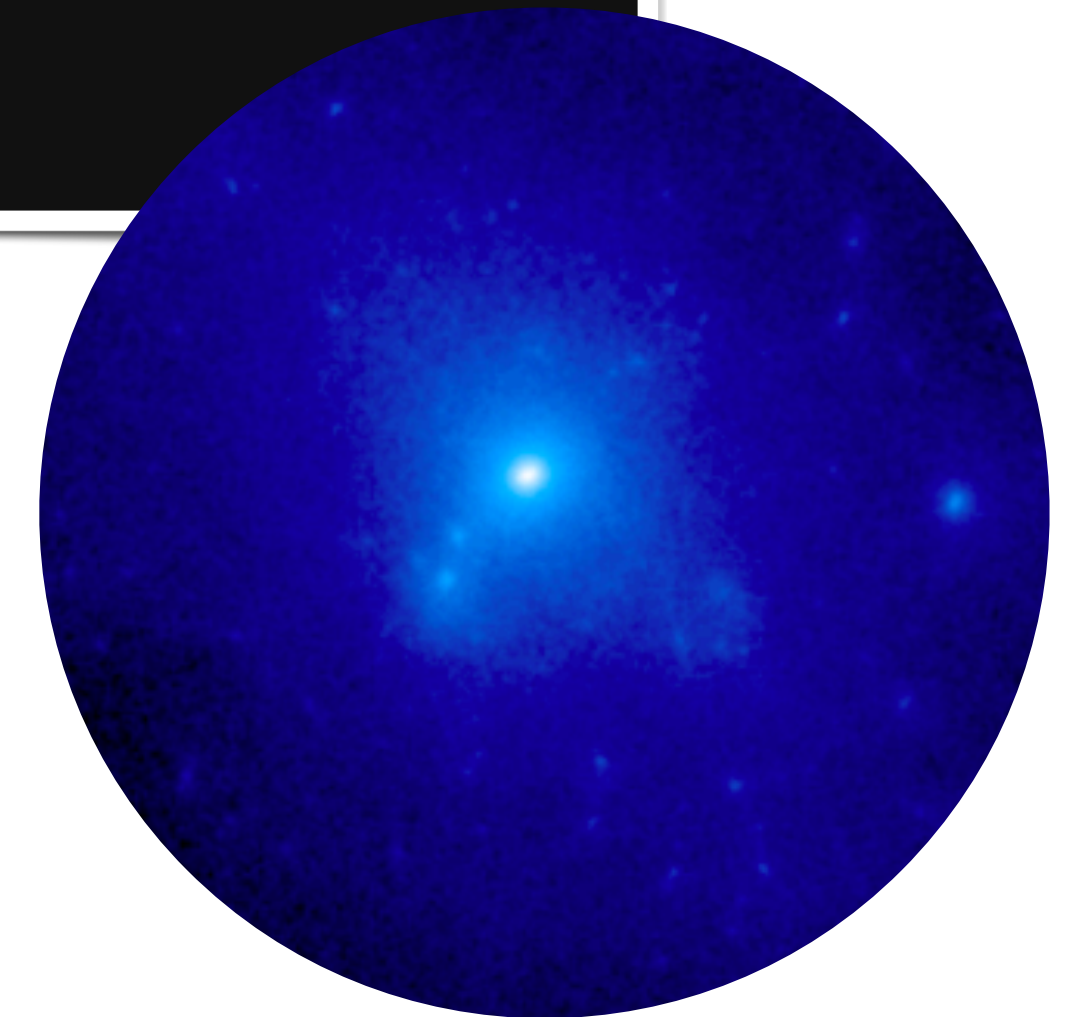
$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + b^2}}$$



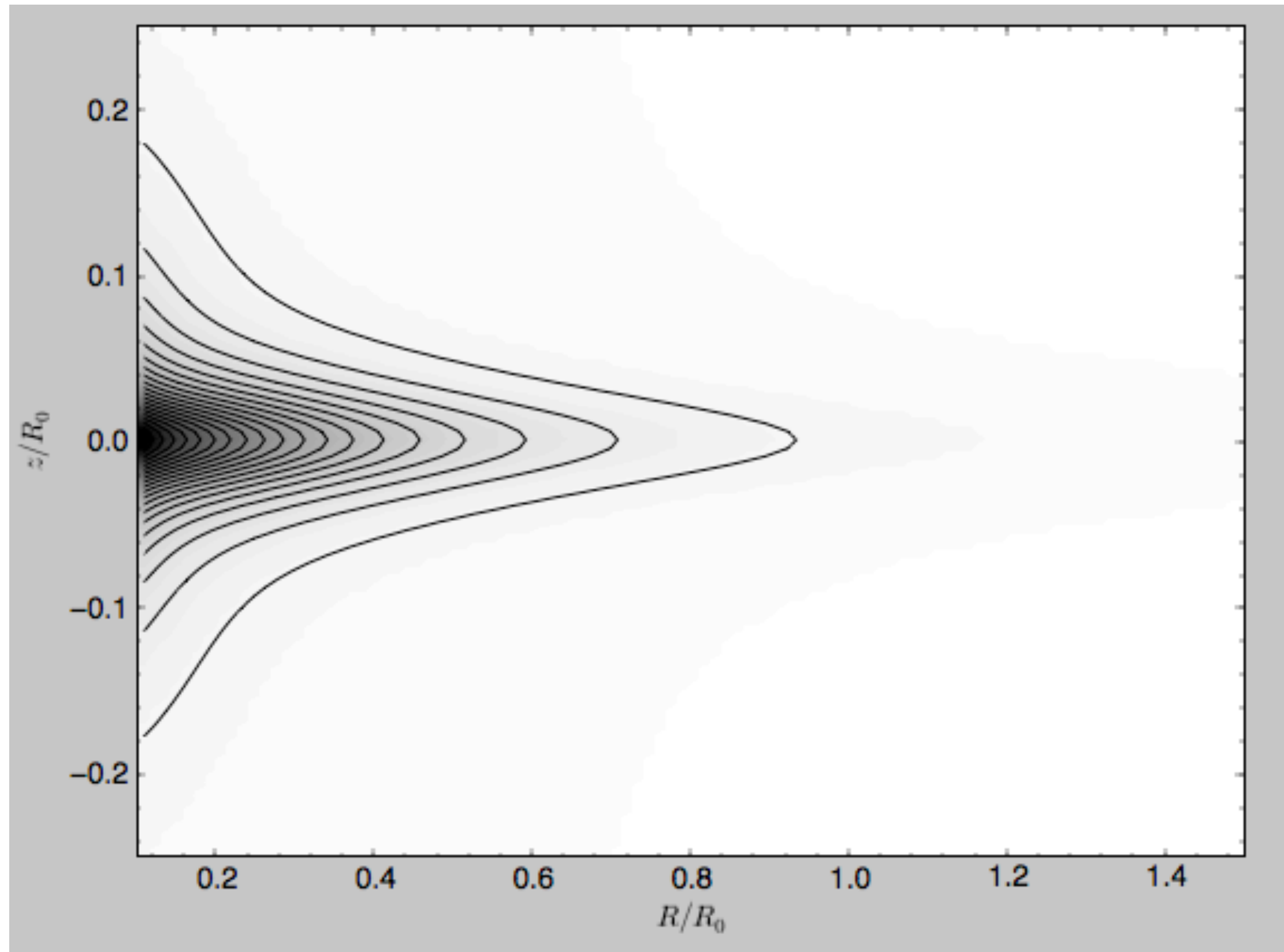
Two-power models

$$\rho(r) = \frac{\rho_0}{(r/r_0)^\alpha (1 + r/r_0)^{\beta-\alpha}}$$

- NFW model (Navarro–Frenk–White): $(\alpha, \beta) = (1, 3)$
- Hernquist model: $(\alpha, \beta) = (1, 4)$
- Jaffe model: $(\alpha, \beta) = (2, 4)$
- Dehnen models: $\beta = 4$



Extensions beyond a spherically symmetric potential:



Miyamoto-Nagai potential

$$\Phi_{\text{MN}}(R, z) = - \frac{G M}{\sqrt{R^2 + (a + \sqrt{z^2 + b^2})^2}}$$

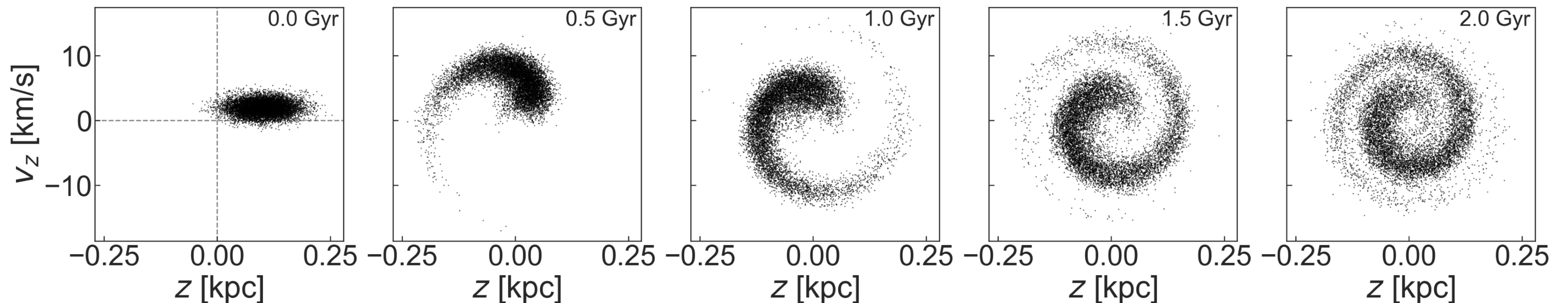
If $a \rightarrow 0$, this reduces to Plummer sphere

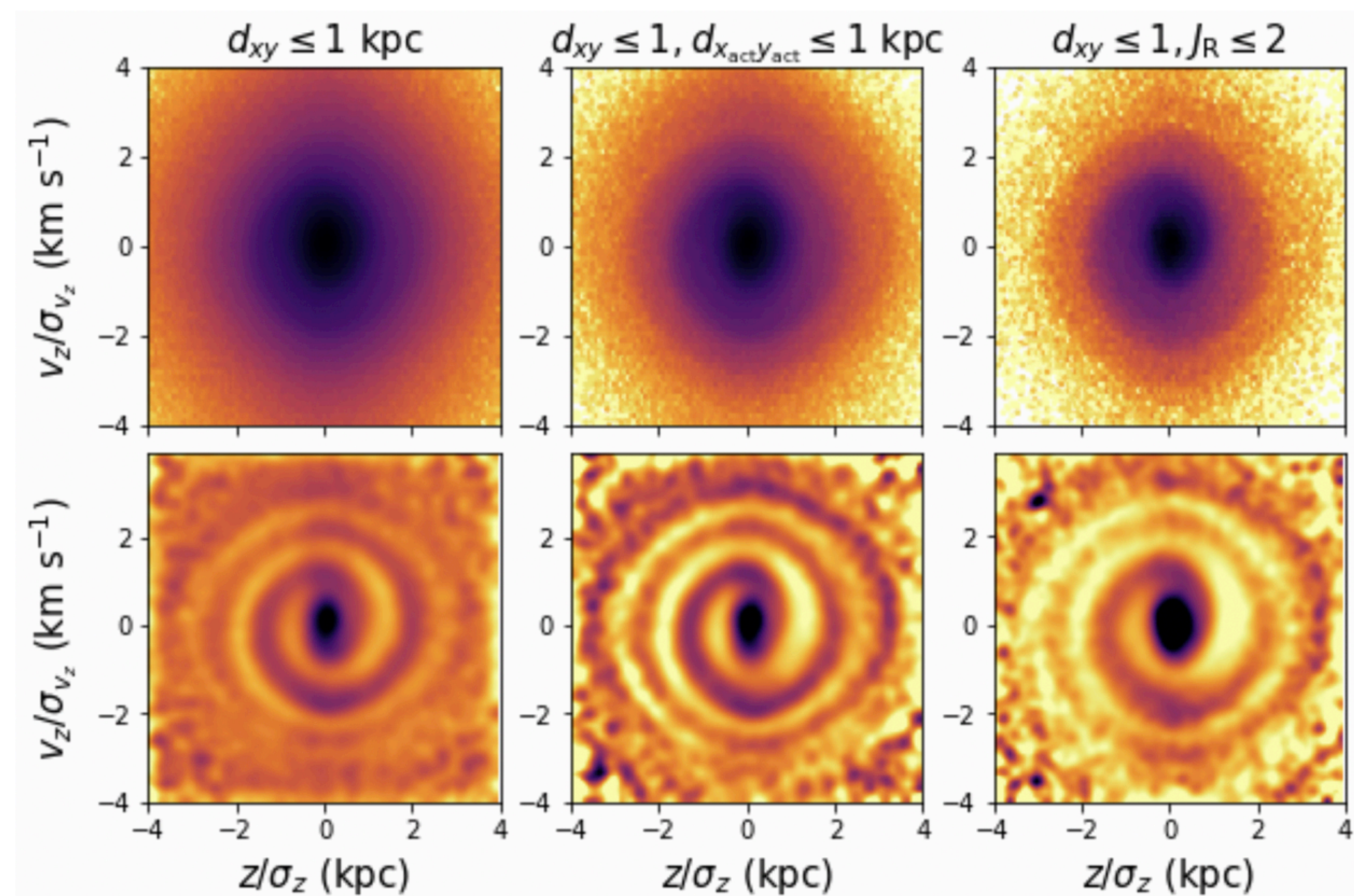
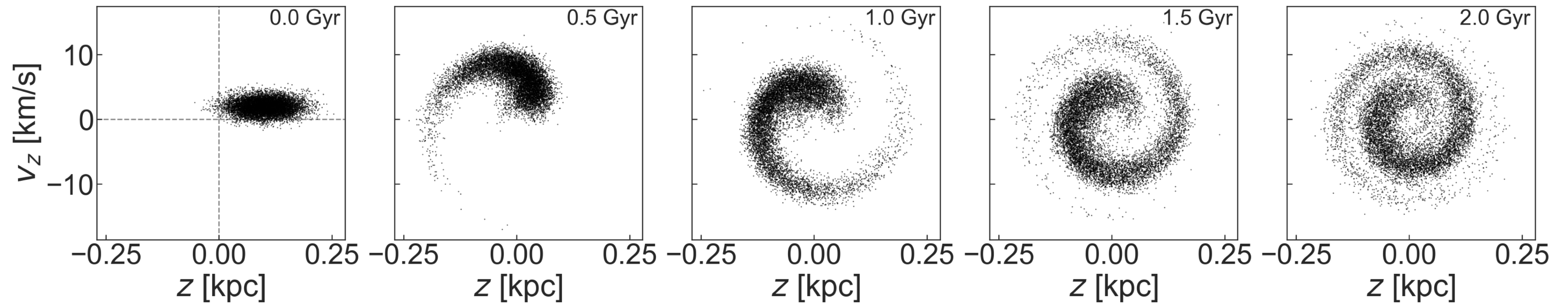
For the MW: $a = 3$ kpc, $b = 0.28$ kpc

Simulating a galaxy

- select initial positions from a density distribution
- assign initial velocities, usually from a velocity dispersion
- integrate the phase space position using a potential

For a Milky Way (MN) potential, you get phase mixing



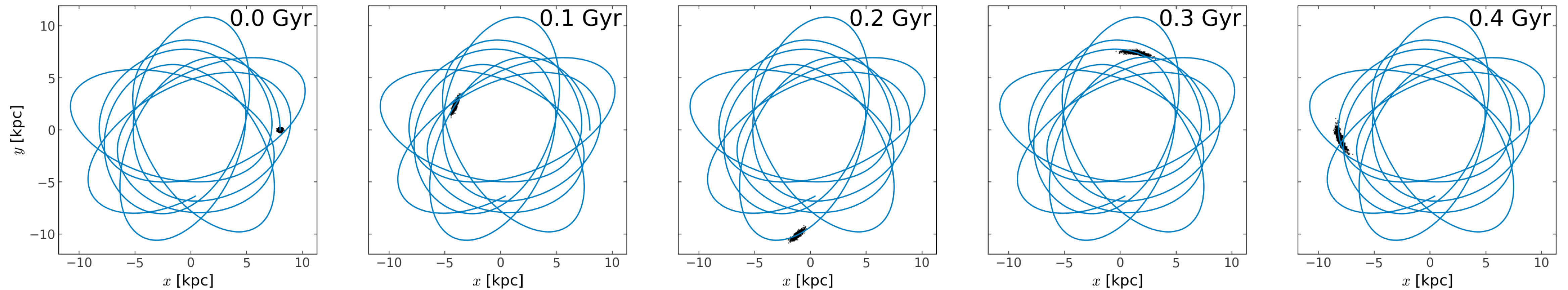


Phase spirals are
observed in Gaia data

Top: number counts

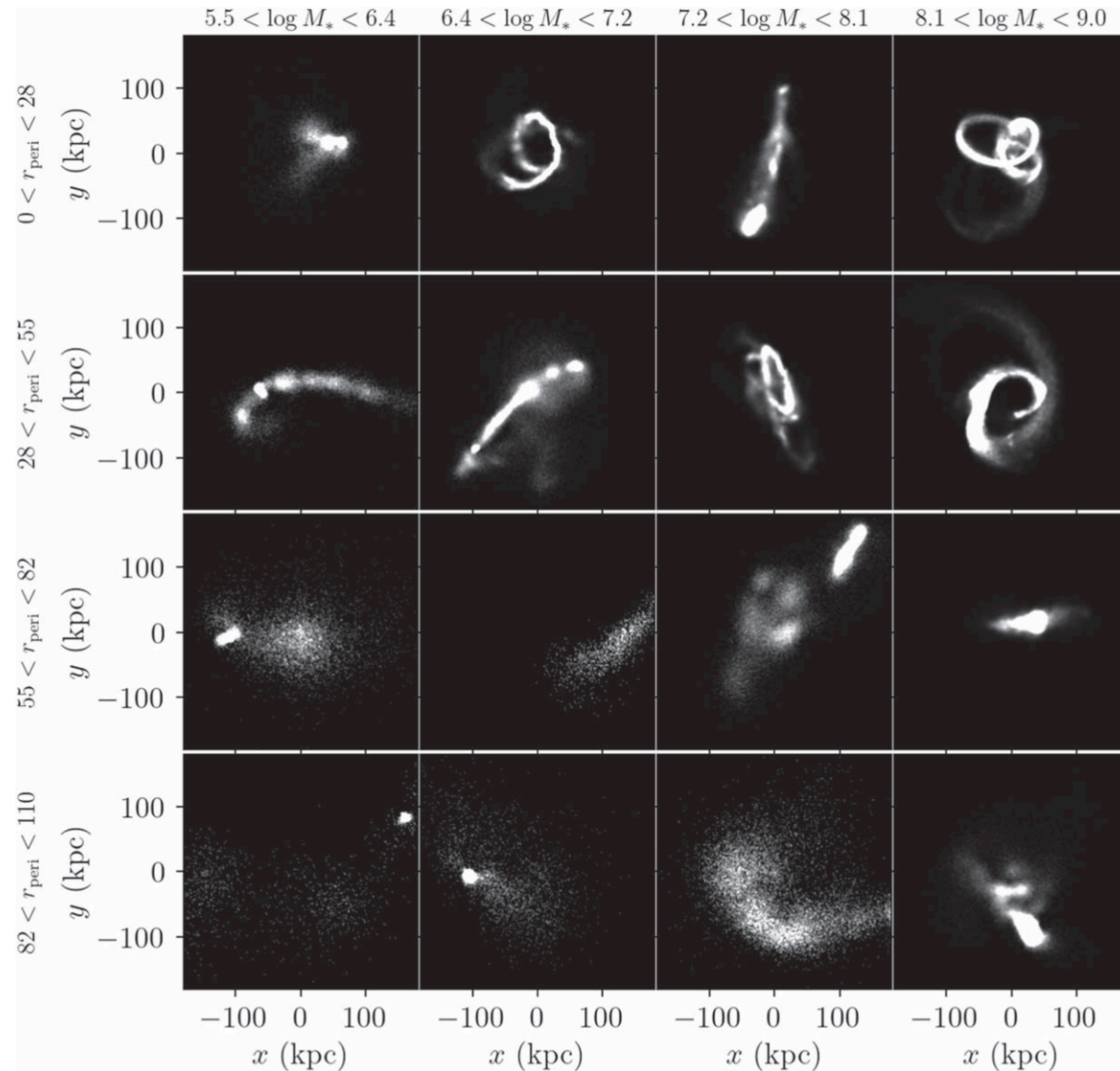
Bottom: smoothed/gaussian subtracted

Tracing the potentials of galaxies



If we initialize a small population of stars on an orbit that will tidally disrupt the system, the stream will elongate along the orbit

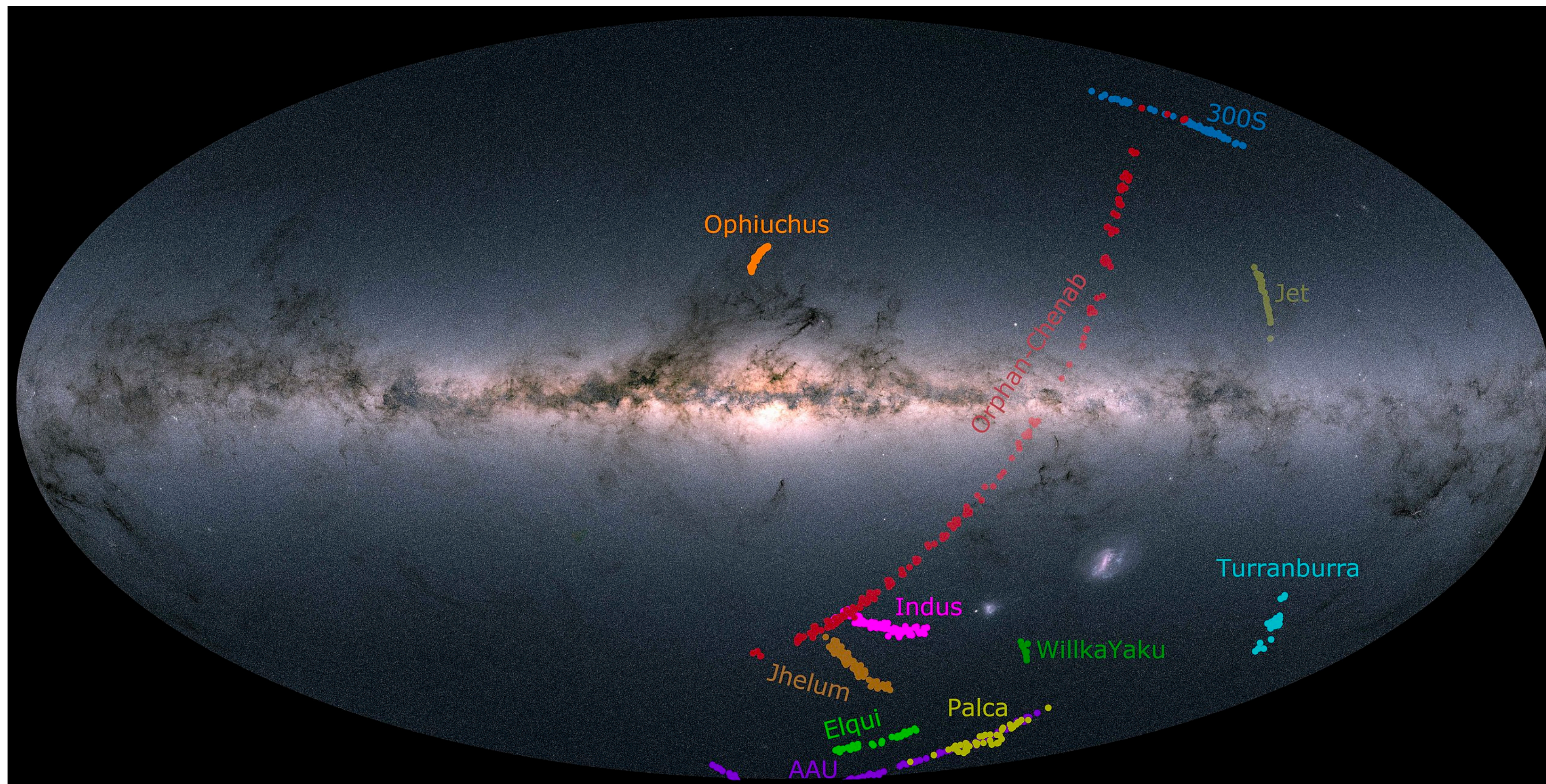
Tracing the potentials of galaxies



Cosmological zoom in simulations show this process happening in many kinds of galaxies

Streams tell you about the galaxy potential

Positions and proper motions from Gaia and radial velocities from spectra allow us to find streams orbiting the Milky Way



Sub-structure in the streams help us study small-scale perturbations like dark matter halos w/ $10^6 - 10^7 M_\odot$

